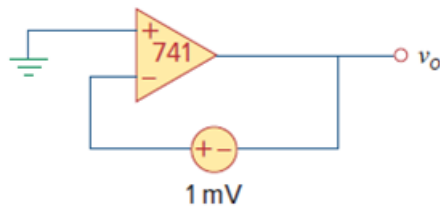


Problem

Using the same parameters for the 741 op amp in Example 5.1, find v_o in the op amp circuit of Fig. 5.45.

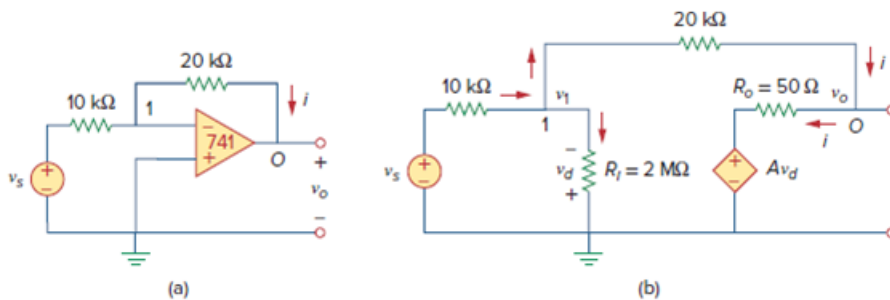
Figure. 5.45:



Example 5.1:

A 741 op amp has an open-loop voltage gain of 2×10^5 , input resistance of $2 \text{ M}\Omega$, and output resistance of 50Ω . The op amp is used in the circuit of Fig. 5.6(a). Find the closed-loop gain v_o/v_s . Determine current i when $v_s = 2 \text{ V}$.

Figure. 5.6: (a) original circuit, (b) the equivalent circuit.



Solution:

Using the op amp model in Fig. 5.4, we obtain the equivalent circuit of Fig. 5.6(a) as shown in Fig. 5.6(b). We now solve the circuit in Fig. 5.6(b) by using nodal analysis. At node 1, KCL gives

$$\frac{v_s - v_1}{10 \times 10^3} = \frac{v_1}{2000 \times 10^3} + \frac{v_1 - v_o}{20 \times 10^3}$$

Multiplying through by 2000×10^3 , we obtain

$$200v_s = 301v_1 - 100v_o$$

Or

$$2v_s \simeq 3v_1 - v_o \Rightarrow v_1 = \frac{2v_s + v_o}{3} \quad (5.1.1)$$

At node O,

$$\frac{v_1 - v_o}{20 \times 10^3} = \frac{v_o - Av_d}{50}$$

But $v_d = -v_1$ and $A = 200,000$. Then

$$v_1 - v_o = 400(v_o + 200,000v_1) \quad (5.1.2)$$

Substituting v_1 from Eq. (5.1.1) into Eq. (5.1.2) gives

$$0 \simeq 26,667,067v_o + 53,333,333v_s \Rightarrow \frac{v_o}{v_s} = -1.9999699$$

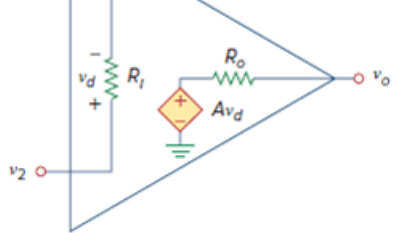
This is closed-loop gain, because the $20\text{-k}\Omega$ feedback resistor closes the loop between the output and input terminals. When $v_s = 2 \text{ V}$, $v_o = -3.9999398 \text{ V}$. From Eq. (5.1.1), we obtain $v_1 = 20.066667 \mu\text{V}$. Thus,

$$i = \frac{v_1 - v_o}{20 \times 10^3} = 0.19999 \text{ mA}$$

It is evident that working with a nonideal op amp is tedious, as we are dealing with very large numbers.

Figure. 5.4: The equivalent circuit of the nonideal op amp.





Step-by-step solution

Step 1 of 5

Refer to circuit diagram in Figure 5.46 in the textbook.

Draw the equivalent circuit of the circuit shown in Figure 5.46.

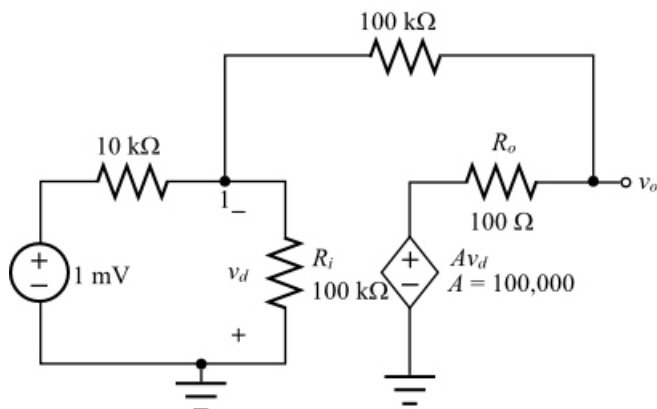


Figure 1

Step 2 of 5

From the circuit shown in Figure 1,

$$v_1 = -v_d$$

Apply Kirchhoff's current law at node 1.

$$\begin{aligned} \frac{v_1 - 1 \text{ mV}}{10 \text{ k}\Omega} + \frac{v_1}{100 \text{ k}\Omega} + \frac{v_1 - v_o}{100 \text{ k}\Omega} &= 0 \\ \frac{v_1}{10 \text{ k}\Omega} + \frac{v_1}{100 \text{ k}\Omega} + \frac{v_1}{100 \text{ k}\Omega} - \frac{v_o}{100 \text{ k}\Omega} &= \frac{1 \text{ mV}}{10 \text{ k}\Omega} \\ \left(\frac{1}{10 \text{ k}\Omega} + \frac{1}{100 \text{ k}\Omega} + \frac{1}{100 \text{ k}\Omega} \right) v_1 - \frac{v_o}{100 \text{ k}\Omega} &= \frac{1 \text{ mV}}{10 \text{ k}\Omega} \\ \left(\frac{10 + 1 + 1}{100 \text{ k}\Omega} \right) v_1 - \frac{v_o}{100 \text{ k}\Omega} &= \frac{1 \text{ mV}}{10 \text{ k}\Omega} \end{aligned}$$

$$12v_1 - v_o = 10 \text{ mV}$$

Substitute $-v_d$ for v_1 in the equation.

$$-12v_d - v_o = 10 \text{ mV} \dots\dots (1)$$

Step 3 of 5

Apply Kirchhoff's current law at node v_o .

$$\frac{v_o - v_1}{100 \text{ k}\Omega} + \frac{v_o - Av_d}{100 \text{ }\Omega} = 0$$

Substitute $-v_d$ for v_1 in the equation.

$$\frac{v_o + v_d}{100 \text{ k}\Omega} + \frac{v_o - Av_d}{100 \text{ }\Omega} = 0$$

$$\frac{v_o}{100 \text{ k}\Omega} + \frac{v_o}{100 \text{ }\Omega} + \frac{v_d}{100 \text{ k}\Omega} - \frac{100000v_d}{100 \text{ }\Omega} = 0 \quad (A = 100,000)$$

$$\left(\frac{1}{100 \text{ k}\Omega} + \frac{1}{100 \text{ }\Omega} \right) v_o + \left(\frac{1}{100 \text{ k}\Omega} - \frac{100000}{100 \text{ }\Omega} \right) v_d = 0$$

$$\left(\frac{1+1000}{100 \text{ k}\Omega} \right) v_o + \left(\frac{1-10^8}{100 \text{ k}\Omega} \right) v_d = 0$$

$$1001v_o - 99999999v_d = 0$$

$$1001v_o = 99999999v_d$$

$$v_o = 99900.099v_d \dots\dots (2)$$

Step 4 of 5

Recall equation (1).

$$-12v_d - v_o = 10 \text{ mV}$$

Substitute $99900.099v_d$ for v_o in the equation.

$$-12v_d - 99900.099v_d = 10 \text{ mV}$$

$$-99912.099v_d = 10 \text{ mV}$$

$$v_d = -100 \text{ nV}$$

Therefore, the differential voltage, v_d is $\boxed{-100 \text{ nV}}$.

Step 5 of 5

Calculate the output voltage.

Recall equation (2).

$$v_o = 99900.099v_d$$

Substitute -100 nV for v_d in the equation.

$$v_o = (99900.099)(-100 \text{ nV})$$

$$= -9.99 \text{ mV}$$

$$= -10 \text{ mV}$$

Therefore, the output voltage, v_o is $\boxed{-10 \text{ mV}}$.